

Design AI factory infrastructure with 800 VDC power distribution and edge computing for energy-optimized operations in high-density computing environments.

Manufacturing × Energy optimisation × Edge computing · OS125 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
69 is the world moving	68 can we play, can we win	MEDIUM 2 named in the evidence	€979m partial evidence · per year	LATER research and standards activi...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about designing AI factory infrastructure that uses 800 VDC power distribution and edge computing to optimize energy use in high-density computing environments. It targets manufacturers deploying AI workloads at the edge, who need to reduce energy consumption while maintaining low-latency decision-making. The timing is driven by recent industry moves to 800 VDC power and the growing need for energy-efficient edge AI.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.4bn €173m – €8.9bn	€979m €124m – €6.4bn	€27.4m €3.5m – €179m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€586k €586k – €1.6m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Edge computing	10.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

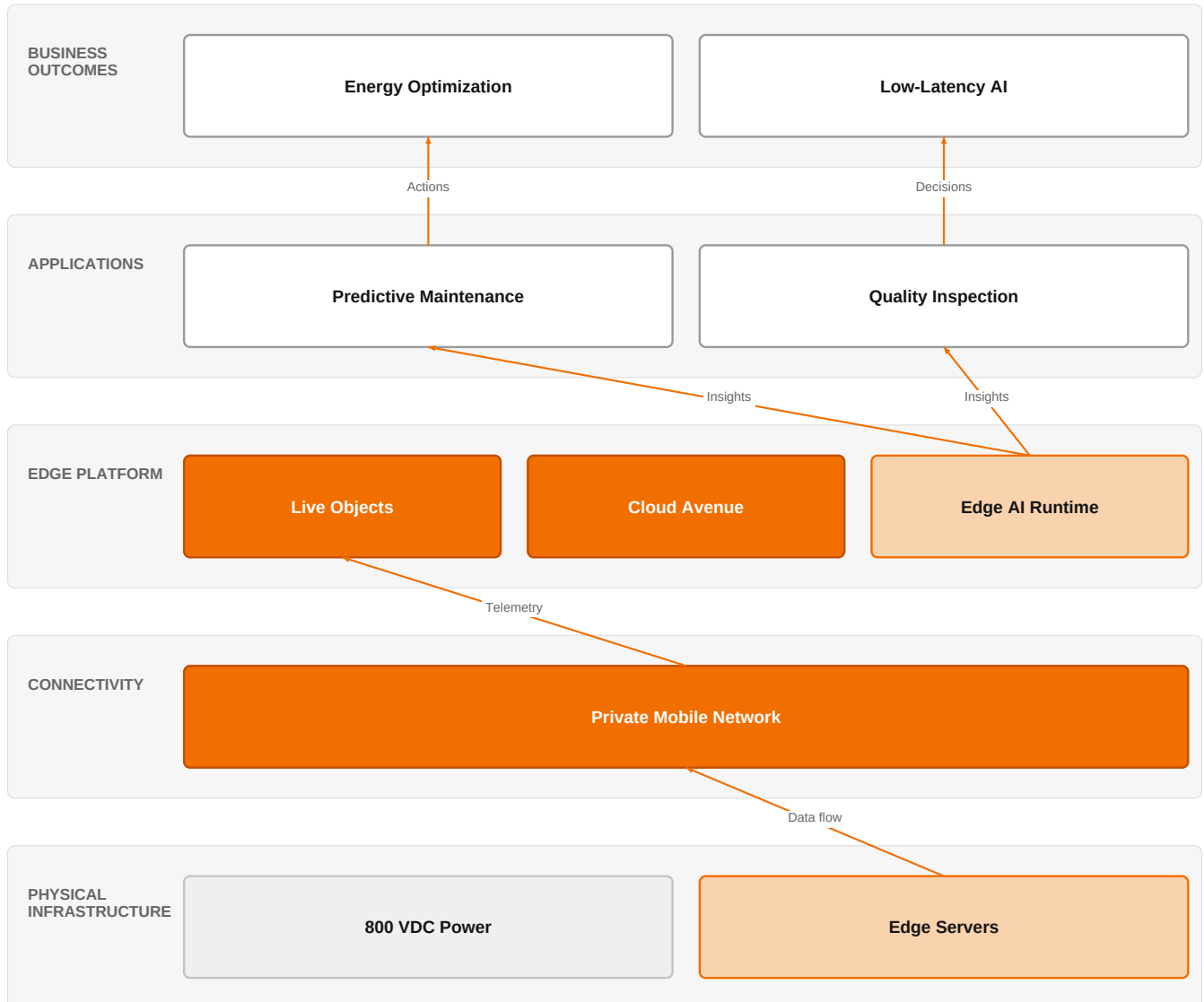
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Energy-Optimized AI Factory Edge



Orange asset Partner Customer-owned Third party / to be sourced

This diagram shows how Orange integrates edge computing, private connectivity, and energy-efficient power to deliver low-latency AI and energy optimization in manufacturing.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The buyer is typically a manufacturing plant operations director or a digital transformation lead, while the pain is felt by plant engineers and IT teams struggling with rising energy costs and the need for real-time AI inference at the edge. The trigger is often a new AI initiative (e.g., predictive maintenance, quality inspection) that requires low-latency processing and cannot rely solely on the cloud, or a mandate to reduce energy consumption across the plant.

Why it is hot now — every claim bound to a dated source

- Schneider Electric and AMD launched the Helios AI Factory Design, focusing on powering next-gen AI infrastructure. [SIG-08272824EFDB]
- Siemens and Reinhausen are engineering solutions to feed 800 VDC to next-gen AI racks, addressing energy demands of AI. [SIG-2AE8872B365A]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble an edge computing solution using its AI/Data/Cloud experts and IoT experts to design and deploy a private mobile network for reliable connectivity, with Live Objects for device management and data ingestion. The solution would leverage Cloud Avenue for secure, sovereign cloud services, and integrate with partners like NVIDIA, Dell, and HPE for edge hardware and AI acceleration. The engagement would start with an assessment of the customer's energy and latency requirements, then design the edge architecture, deploy the infrastructure, and optimize it for energy efficiency.

Why Orange can win

Orange can win by combining its expertise in connectivity, cloud, and IoT with strong security certifications (ISO 27001, TISAX) and a portfolio that includes private mobile networks and sovereign cloud options. The ability to orchestrate a complete edge-to-cloud solution, backed by proven references in manufacturing (e.g., Saint-Gobain Glass, Colgate-Palmolive), gives Orange credibility. However, the specific 800 VDC power distribution expertise is not in Orange's core assets, so Orange would need to partner with specialists like Siemens or Schneider Electric, which could dilute its differentiation.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer addresses the use case but not this technology	unconfirmed
Cloud Avenue / Ready SecNumCloud	offer	L1	offer provides the technology in a matching domain	unconfirmed
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 6.9; 2 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Siemens is strong in industrial OT and is actively working on 800 VDC power for AI racks, making them a direct competitor for the power distribution aspect. Schneider Electric also sells edge computing and is active in manufacturing, with a focus on energy management. Huawei competes in connectivity solutions and has partnered with AIS for smart manufacturing. AWS, Deutsche Telekom, Lumen, NTT DATA, and HPE all sell edge computing and could offer similar solutions, with hyperscalers like AWS strong in cloud integration and telecom peers strong in connectivity.

Sources: SIG-2AE8872B365A theregister.com 2026-08-17; SIG-2FF6CAD27F35 developingtelecoms.com 2026-08-06

Competitor	Category	Position here	Basis	Evidence
Siemens	Industrial / OT specialist	active in Manufacturing	evidenced	SIG-2AE8872B365A
Huawei	Network equipment vendor	competes in Connectivity Solutions — also an Orange partner	evidenced	SIG-2FF6CAD27F35
AWS	Hyperscaler	sells Edge computing; competes in Cloud — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; active in Manufacturing; competes in Cloud, Connectivity Solutions	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Cloud, Connectivity Solutions	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing; active in Manufacturing	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions — also an Orange partner	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is the current power architecture in your high-density computing areas, and have you evaluated 800 VDC as an option?
- 2 Which AI workloads are you planning to run at the edge, and what are your latency and energy consumption targets?
- 3 How are you currently managing energy consumption across your plant, and what is the biggest pain point?
- 4 Do you have a private network or are you relying on public connectivity for your edge devices?
- 5 What is your timeline for deploying new AI capabilities, and is there a specific production line or process that is the priority?

Objections you will meet

They will say	You can answer
800 VDC is too new and unproven for our operations.	It's true that 800 VDC is still emerging, but industry players like Siemens and Schneider Electric are already investing in it. We can start with a pilot on a non-critical line to validate the benefits before scaling.
We already have a cloud strategy with a hyperscaler; why do we need edge?	Edge is complementary to cloud, especially for low-latency and energy-sensitive applications. Our solution integrates with your existing cloud, bringing processing closer to the data source to reduce latency and energy use.
We're concerned about the total cost of ownership.	While the initial investment may be higher, the energy savings and operational efficiencies from edge AI can offset costs over time. We can model the ROI based on your specific workloads and energy prices.

Risks and unknowns

The main risk is that 800 VDC power distribution is still emerging, with limited standards and proven deployments, which could slow adoption. Additionally, Orange lacks in-house expertise in high-voltage power distribution, so it would rely on partners, potentially reducing margins. It is also unknown how quickly manufacturers will adopt this technology versus sticking with traditional AC power, and whether the energy savings justify the infrastructure investment.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Develop a bid brief for a manufacturing prospect, highlighting Orange's Live Objects and Cloud Avenue offers, and how they can support energy-optimized edge computing, using Saint-Gobain Glass as a reference for industrial IoT deployments.

Sales	In a conversation with a manufacturing client's operations or IT leader, mention Orange's expertise in IoT and edge computing, referencing successful deployments at Colgate-Palmolive and Mondelez International, and ask about their energy optimization challenges in high-density computing.
Strategist / Innovator	Commission a deep-dive brief on 800 VDC power distribution and edge computing for AI factories, leveraging Orange's AI/Data/Cloud experts and IoT experts to assess feasibility and market readiness.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AEA954F39CAC	2026-04-01	cordis.europa.eu	1	HIGH SPEED AI-DRIVEN ENGINEERING SIMULATIONS FOR ADVANCED PRODUCT DEVELOPMENT
SIG-9CFA159A98D6	2026-01-06	openalex.org	1	Roadmap on Artificial IntelligenceAugmented Additive Manufacturing
SIG-3772E07403F3	2026-01-02	openalex.org	1	Cloudburst AI: Orchestrating SRE-Driven AI Inference Across Multi-Cloud and Edge Zones
SIG-8D00014E2BA4	2026-01-01	Lecture Notes in Networks...	1	Edge Artificial Intelligence for Low-Latency Decision-Making in Intelligent Manufacturing Syste...
SIG-D8ADED8BAD839	2026-01-01	Elsevier BV	1	Robust Energy–Latency Co-Optimization for Power Allocation and Computation Offloading in Edge C...
SIG-0964410DC575	2025-12-17	openalex.org	1	Generative and Predictive AI for digital twin systems in manufacturing
SIG-0860D0B1EBA0	2025-12-05	openalex.org	1	Deploying TinyML for energy-efficient object detection and communication in low-power edge AI s...
SIG-04BD044A1960	2025-11-18	openalex.org	1	Cognitive Edge Computing: A Comprehensive Survey on Optimizing Large Models and AI Agents for P...
SIG-2AE8872B365A	2026-08-17	theregister.com	2	Siemens and Reinhausen turn up the voltage for hungry AI racks
SIG-32E1C57B5BBF	2026-08-17	automationworld.co m	2	Beyond AI Pilots : How Systemic AI Drives Scalable Industrial Automation Across Manufacturing P...
SIG-08272824EFDB	2026-08-13	miragenews.com	2	Schneider , AMD Launch Helios AI Factory Design
SIG-2FF6CAD27F35	2026-08-06	developingtelecoms. com	2	AIS and Huawei Cloud team for smart manufacturing powered by 5G and AI
SIG-D29A41589B0F	2026-07-25	arxiv.org	3	Movable-Antenna Assisted Energy Minimization in UAV-Enabled Mobile Edge Computing Systems
SIG-5ADA6B03AA90	2026-05-05	arxiv.org	3	Physics-Grounded Multi-Agent Architecture for Traceable, Risk-Aware Human-AI Decision Support i...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: what_is_changing (named entities that were never supplied: AMD)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:15+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.